Out of the Hyperintensional Bungalow

Dean McHugh*

October 2026 – Comments welcome!

Abstract

This paper argues that there is an important but overlooked distinction in the realm of hyperintensionality. Some hyperintensional phenomena are due to differences in truthmakers, others due to differences in alternatives. I show that existing approaches—most notably, those based on truthmaker and alternative semantics—fail to respect this distinction, leading to incorrect results concerning the hyperintensional behaviour of permission and conditional antecedents. While each framework in isolation fails, this paper synthesises them into a new framework that respects the truthmaker–alternative distinction and delivers the right results. Where alternative and truthmaker semantics are each hyperintensional 'bungalows', having only one level of hyperintensional content, I show that the proper treatment of permission and conditional antecedents calls for two. It is time to leave our separate bungalows and upgrade to a two-storey house.

10,967 words

1 Introduction

Within philosophical logic and semantics, it is often thought the meaning of a statement is determined by the possible worlds in which it is true—a tradition known as *possible worlds semantics* (Lewis 1970, Partee 1989). According to the standard theory of modality, for example, a statement like 'Eve may sing' is true just in case there is some deontically ideal world where Eve sings (Hanson 1965, Kratzer 1977).[1] And on the influential Stalnaker–Lewis analysis of counterfactuals, *if A, would C* is true just in case C holds at the closest worlds where A is true (Stalnaker 1968, Lewis 1973).

Possible worlds semantics for permission and conditionals satisfies the following principles, respectively, where $\Diamond$ denotes 'it is permitted that' and $A > C$ denotes *if A, would C*.[2]

Modal Substitution. If A and B are true in all the same worlds, then so are $\Diamond A$ and $\Diamond B$.
Conditional Substitution. If A and B are true in all the same worlds, then so are $A > C$ and $B > C$.

*I am grateful to Kit Fine, Lucas Champollion, Daniel Lassiter, Richard Roth, and Christopher Sun for helpful discussions of this paper. This work was completed during a Fulbright Scholarship at New York University, supervised by Kit Fine, and during the Communicating Causality project (AH/Z507490/1) at the University of Edinburgh, supervised by Daniel Lassiter.

[1] Portner (2009:47) and Chrisman (2015:69–73) call this the 'standard' theory of deontic modals.

[2] Following standard practice, I assume that a wide variety of constructions express $\Diamond$, including 'it is allowed that A', and, where A is of the form Fa, 'a may F' and 'a is permitted to F'.

This paper examines two families of counterexample to these principles. The first, due to Kit Fine (2014), concerns the behaviour of *infinitely many*. The second concerns the differences between disjunction and other logical vocabulary. I argue that these counterexamples provide distinct sources of hyperintensional behaviour in natural language—a distinction overlooked by extant frameworks.

I show that a wide range of extant semantic frameworks fail to capture both kinds of counterexample simultaneously. The approaches I consider fall into two broad families: those based on truthmaker semantics and those based on alternative semantics. In response, I propose a novel synthesis of truthmaker and alternative semantics. In alternative semantics, the meaning of a statement is a set of propositions, called its *alternatives*, where each proposition is a set of possible worlds. I propose to rethink alternatives not as sets of worlds but as sets of truthmakers. The resulting framework I call *alternative truthmaker semantics*.

Thus where alternative semantics and truthmaker semantics are 'bungalows', with one level of hyperintensional content (alternatives or truthmakers in isolation), permission and conditionals call for two. This framework, I argue, inherits the best of both underlying frameworks. From truthmaker semantics we take the ability to capture the hyperintensional character of *infinitely many*. From alternative semantics we take its treatment of disjunction. The overall lesson is that there are many roads to hyperintensionality, even within a single environment such as permission or conditional antecedents. We need a framework that respects their differences.

The plan. The remainder of this section presents the two challenges we aim to account for. Section 2 argues against truthmaker semantics of modals and conditionals. In Section 3 I raise two objections against my argument and offer replies. Section 4 presents some challenges for accounts of modals and conditionals within alternative semantics. My positive proposal comes in Section 5, showing how the combination of truthmaker and alternative semantics successfully accounts for the data under consideration. Section 6 concludes.

1.1 Challenge 1: Trouble in Paradise

David Lewis once wrote that logical space—the set of possible worlds—'is a paradise for philosophers' (1986:4). We now have ample evidence that this paradise, like the Garden of Eden, was not to last. One piece of evidence comes from Kit Fine (2014). He asks us to imagine Eve in Alternative Eden, containing the Forbidden Fruit a_0 in addition to infinitely many other apples $a_1, a_2, \ldots$. God tells Eve:

(1) You may eat infinitely many of the apples $a_1, a_2, \ldots$.

Intuitively, this does not permit Eve to eat the Forbidden Fruit a_0. Suppose God had instead said:

(2) You may eat infinitely many of the apples $a_0, a_1, a_2, \ldots$.

In contrast, this intuitively does permit Eve to eat a_0 alongside infinitely many apples.[3]

[3]Note that the contrast does not depend on this particular formulation of the example. As Fine (2014:329) observes, it persists under a wide range of formulations. For example, suppose that the Forbidden Fruit a_0 is red, while the safe apples $a_1, a_2, \ldots$ are green. Compare:

(i) a. You may eat infinitely many of the green apples.

Fine (2014:328) observes that conditionals exhibit an analogous contrast.[4]

(3) a. If Eve ate infinitely many of the apples $a_1, a_2, \ldots$, she would be OK.
b. If Eve ate infinitely many of the apples $a_0, a_1, a_2, \ldots$, she would be OK.

The first is intuitively true but the second is not.

The puzzle is that the embedded statements are necessarily equivalent.

(4) a. Eve eats infinitely many of the apples $a_1, a_2, \ldots$.
b. Eve eats infinitely many of the apples $a_0, a_1, a_2, \ldots$.

If one is true, the other must be too, for a difference of one cannot separate the finite from the infinite.[5] Nonetheless, they differ in what *makes* them true: the state of Eve eating a_0 alongside infinitely many of the other apples verifies (4b) but not (4a). Modal Substitution predicts God's statements in (1) and (2) to be equivalent, and Conditional Substitution predicts the conditionals in (3) to be equivalent.

1.2 Challenge 2: The Function of Disjunction

The second challenge to Modal and Conditional Substitution comes from the relationship between disjunction and other logical vocabulary. Consider first permission. Suppose God instructs Adam that he may pick at most five pomegranates from the Garden of Eden. Consider:

(5) a. Adam may pick more than four pomegranates.
b. Adam may pick five or more pomegranates.

Intuitively, Adam may pick more than four pomegranates, since he may pick five. But it is wrong to say that he may pick five or more, since he may not pick more than five. The embedded statements, however, are true in all the same worlds: Adam picks more than four if and only if he picks five or more.[6]

---

b. You may eat infinitely many of the apples.

The contrast persists: only the second permits Eve to eat the red apple alongside infinitely many green apples.

[4] Goodsell (2022) presents a similar case.

[5] One may worry that (4a) and (4b) are not necessarily equivalent for the following reason. Take a world where a_0 does not exist and Eve eats infinitely many of $a_1, a_2, \ldots$. In such a world, (4a) is clearly true, but one might think (4b)'s truth value is undefined due to presupposition failure, owing to the non-referring term a_0, similar to 'The King of France is not bald' (Strawson 1950). In response, consider sentences explicitly stating that $a_0, a_1, a_2, \ldots$ exist.

(i) a. You may look at every apple $a_0, a_1, \ldots$ and eat infinitely many of $a_1, a_2, \ldots$.
b. You may look at every apple $a_0, a_1, \ldots$ and eat infinitely many of $a_0, a_1, a_2, \ldots$.

(ii) a. If there were infinitely many apples $a_0, a_1, \ldots$ and Eve ate infinitely many of $a_1, a_2, \ldots$, she'd be OK.
b. If there were infinitely many apples $a_0, a_1, \ldots$ and Eve ate infinitely many of $a_0, a_1, a_2, \ldots$, she'd be OK.

The embedded sentences no longer suffer from the alleged presupposition failure, but the contrast persists.

[6] Note that the example wouldn't work for an incremental verb such as *eat*. One who eats four and a half pomegranates has eaten more than four but not five or more.

A second example comes from De Morgan's Law.

(6) a. You have to work Saturday, but you don't have to work both Saturday and Sunday.
$\Box A \wedge \neg\Box(A \wedge B)$

b. You have to work Saturday, but you may take off either Saturday or Sunday.
$\Box A \wedge \Diamond(\neg A \vee \neg B)$

These elicit a striking contrast. The first is fine but the second seems internally inconsistent. Permission to take off either Saturday or Sunday implies permission to take off Saturday, contradicting the obligation to work Saturday.[7] This is a counterexample to Modal Substitution assuming De Morgan's Law (the equivalence of $\neg(A \wedge B)$ and $\neg A \vee \neg B$) and Duality (the equivalence of $\neg\Box A$ and $\Diamond\neg A$).

However, many reject Duality.[8] To sidestep the issue, we may rephrase the counterexample purely in terms of permission.[9]

(7) a. You have to work Saturday, but you are permitted not to work both Saturday and Sunday.
$\Box A \wedge \Diamond\neg(A \wedge B)$

b. You have to work Saturday, but you are permitted to take off either Saturday or Sunday.
$\Box A \wedge \Diamond(\neg A \vee \neg B)$

This, while more cumbersome than (6), elicits the same contrast. The first is consistent while the second is contradictory.

Ciardelli, Zhang and Champollion (2018) present evidence that conditional antecedents also break De Morgan's Law: $\neg(A \wedge B) > C$ is not equivalent to $(\neg A \vee \neg B) > C$. In their scenario, there are two switches, A and B, connected to a light. Each switch can be up or down. The light is on just in case the switches agree, i.e. are either both up or both down. Currently, both switches are up, so the light is on. Compare:

(8) a. If switch A or switch B was down, the light would be off. $(\overline{A} \vee \overline{B}) > \textsc{Off}$

b. If switch A and switch B were not both up, the light would be off. $\neg(A \wedge B) > \textsc{Off}$

The antecedents form an instance of De Morgan's Law on the natural assumption that for each switch, being down is equivalent to being not up ($\overline{A} \equiv \neg A$ and $\overline{B} \equiv \neg B$). Intuitively, however, the first is true but the second is not, a contrast Ciardelli, Zhang and Champollion observed in their experiment: a majority of

[7] The inference from $\neg\Box(A \wedge B)$ to $\neg\Box A \wedge \neg\Box B$ is known as *negative free choice*. Marty et al. (2021) found that while the inference is sometimes drawn, it is far less robust than the free choice inference from $\Diamond(A \vee B)$ to $\Diamond A \wedge \Diamond B$. In light of these data, they propose that free choice is an entailment but negative free choice is an implicature.

[8] Examples include Fusco (2015), Anglberger, Gratzl and Roy (2015), Anglberger, Korbmacher and Faroldi (2016) and Fine (2018:649–50). Another tradition—represented by von Wright (1963), Lokhorst (1997), Asher and Bonevac (2005) and Barker (2010), among others—distinguishes between strong and weak permission, with the strong form requiring explicit permission and the weak form mere absence of prohibition. Duality then fails if $\Diamond$ expresses strong permission. See Hansson (2013) for a useful overview. Some recent theories that go beyond possible worlds semantics but nonetheless accept both Duality and De Morgan's Law are Aloni (2018) and Willer (2018).

[9] Note that 'you are permitted not to work both Saturday and Sunday' and 'you are permitted to take off either Saturday or Sunday' exhibit a scope ambiguity, and may also be read as $\Diamond\neg A \wedge \Diamond\neg B$ and $\Diamond\neg A \vee \Diamond\neg B$, respectively. Here we are concerned with their respective readings $\Diamond\neg(A \wedge B)$ and $\Diamond(\neg A \vee \neg B)$.

participants (69%) judged (8a) true but only a small minority (22%) judged (8b) true.

2 Truthmaker Theories

Let us see how truthmaker theories of modals and conditionals fare with the above challenges.

Fine (2014) uses Alternative Eden to motivate a truthmaker semantics of permission. For while (4a) and (4b) are true in the same worlds, they are naturally assigned different truthmakers; in particular, on Fine's regimentation, the state of Eve eating the Forbidden Fruit a_0 alongside infinitely many apples verifies the latter but not the former.[10]

This allows truthmaker semantics of permission and counterfactuals to distinguish the embedded sentences in (4). Extant truthmaker semantics of permission (Fine 2014, 2018, Anglberger, Korbmacher and Faroldi 2016) do so by satisfying the following principle.[11]

> *Universal Realisability of Permission.*
> $\Diamond A$ is true iff every truthmaker for A is permitted.

This distinguishes between God's statements (1) and (2), predicting the first to be true and the second to be false.

Extant truthmaker semantics of conditionals (Fine 2012a,b, Briggs 2012, Santorio 2018, Rosella and Sprenger 2024, Brast-McKie 2025) satisfy an analogous principle.[12]

> *Universal Realisability of the Antecedent.*
> $A > C$ is true iff every truthmaker of A leads to the truth of C.

This also distinguishes between the conditionals in (3), predicting (3a) to be true and (3b) to be false.

---

[10] Formally, for each apple a, we let p_a be the atomic sentence that Eve eats apple a, whose sole verifier is s_a. Fine analyses the sentences in terms of infinitary conjunction and disjunction, as $\bigvee_{X \subseteq^{\infty} \{a_1, a_2, \ldots\}} p_X$ and $\bigvee_{X \subseteq^{\infty} \{a_0, a_1, a_2, \ldots\}} p_X$, respectively, where p_X denotes $\bigwedge_{a \in X} p_a$ and $X \subseteq^{\infty} Y$ that X is an infinite subset of Y. Thus 'Eve may eat infinitely many of $a_0, a_1, \ldots$' expresses $\Diamond(\bigvee_{X \subseteq^{\infty} \{a_0, a_1, \ldots\}} p_X)$ and 'Eve may eat infinitely many of $a_1, \ldots$' expresses $\Diamond(\bigvee_{X \subseteq^{\infty} \{a_1, \ldots\}} p_X)$. Note that 'verifier' is synonymous with 'truthmaker'; I use them interchangeably.

[11] These theories differ on the details but all share the principle's universal requirement—our target in what follows. Fine (2014) assumes a primitive set of permitted states and takes $\Diamond A$ to be true iff *every* truthmaker for A is permitted. Fine (2018:639) designates a set of ideal actions (what he calls a 'code of conduct') and proposes that an action X is permitted iff *every* truthmaker for X is part of an ideal action. From this we may also extract a set of permitted states; namely, those that are part of an ideal action. Fine also introduces the notion of weak permission, whereby X is weakly permitted iff *every* truthmaker for X is compatible with an ideal action (Fine 2018:642). Anglberger, Korbmacher and Faroldi (2016) associate with each action a set of actions that it renders 'admissible', and take an act to verify $\Diamond A$ iff it renders admissible *every* truthmaker for A.

[12] As with permission, different theories cash out the principle in different ways but share its universal requirement. For Fine (2012a,b), $A > C$ is true iff for every truthmaker s of A, every possible outcome of s contains a truthmaker of C; Brast-McKie (2025) provides an analysis of Fine's possible outcome relation, retaining the overall shape of Fine's semantic clause. For Briggs (2012), supposition is represented by intervention in a structural causal model. Santorio (2018) provides his own notion of truthmaking (discussed below) and takes $A > C$ to be true (on its distributive reading) iff for every truthmaker p of A, C is true at the closest p-worlds. On Rosella and Sprenger's (2024) probabilistic account, we interpret 's leads to the truth of C' to mean that intervening to make s true results in C having probability 1.

2.1 A Problem for Truthmaker Theories: De Morgan's Law

While truthmaker theories of permission and conditionals account for Challenge 1, they struggle with Challenge 2, concerning the behaviour of disjunction. Fine's (2017b) truthmaker clauses, on both the bilateral and unilateral variants, predict $\neg(A \wedge B)$ and $\neg A \vee \neg B$ to have the same truthmakers. Given this, truthmaker semantics of permission and conditionals fail to distinguish between the sentences in (7) and (8), respectively, an undesirable result.

The problem is not unique to Fine's clauses. Santorio (2018) proposes a truthmaker semantics of conditionals that satisfies Universal Realisability of the Antecedent, but offers an alternative definition of truthmakers. However, Santorio's algorithm also predicts $\neg(A \wedge B)$ and $\neg A \vee \neg B$ to have the same truthmakers. I relegate the details to a footnote for the interested reader.[13] Santorio's notion of truthmakers therefore also fails to capture the failure of De Morgan's Law in (8).

To account for the failure of De Morgan's Law in conditional antecedents, some have proposed modifying the truthmaker clauses, designing systems where the truthmakers of $\neg(A \wedge B)$ include not only those of $\neg A \vee \neg B$ but also those of $\neg A \wedge \neg B$ (Bernard and Champollion 2018, Champollion and Bernard 2024, Li 2023, De Rizzo 2025). This successfully accounts for the De Morgan violations for conditionals. However, plugging in these truthmakers would not capture the failure of De Morgan's Law for permission. The original truthmaker clauses and the proposed modifications both agree that every truthmaker for $\neg A$ or $\neg B$ is a truthmaker for $\neg(A \wedge B)$.[14] Universal Realisability of Permission therefore predicts $\Diamond\neg(A \wedge B)$ to entail $\Diamond\neg A$ and $\Diamond\neg B$. As (7) shows, this is too strong. It predicts that permission to not work both days

[13] Santorio builds on Katzir's (2007) structural theory of alternatives, according to which a sentence S' counts as an alternative to a sentence S iff S' is contextually relevant and no more syntactically complex than S. The contextually relevant alternatives are those whose items have been pronounced in the context and result from substituting items from the lexicon (e.g. swapping $\wedge$ for $\vee$). For De Morgan's Law, Katzir's procedure returns the following alternative sets.

(i) a. $Alt_{\neg(A \wedge B)} = \{\neg(A \wedge B), \neg(A \vee B), A \wedge B, A \vee B, \neg A, \neg B, A, B\}$
b. $Alt_{\neg A \vee \neg B} = \{\neg A \vee \neg B, \neg A \wedge \neg B, A \vee \neg B, A \wedge \neg B, \neg A \vee B, \neg A \wedge B, A \vee B, A \wedge B, \neg A, \neg B, A, B\}$

As the reader may verify, Santorio's algorithm predicts $\neg(A \wedge B)$ and $\neg A \vee \neg B$ to have the same truthmakers; namely, $|A \wedge \neg B|$, $|\neg A \wedge B|$, and $|\neg A \wedge \neg B|$, where $|S|$ is the set of worlds where sentence S is true.

This result is robust across other choices of alternatives. Consider the Sauerland (2004)-style alternatives that result from deletion and optionally swapping conjunction and disjunction, without removing negation.

(ii) a. $Alt_{\neg(A \wedge B)} = \{\neg(A \wedge B), \neg(A \vee B), \neg A, \neg B\}$
b. $Alt_{\neg A \vee \neg B} = \{\neg A \vee \neg B, \neg A \wedge \neg B, \neg A, \neg B\}$

Consider also the result of closing the Sauerland-alternatives under negation removal (if $\neg S \in Alt$, then $S \in Alt$).

(iii) a. $Alt_{\neg(A \wedge B)} = \{\neg(A \wedge B), \neg(A \vee B), \neg A, \neg B, A \wedge B, A \vee B, A, B\}$
b. $Alt_{\neg A \vee \neg B} = \{\neg A \vee \neg B, \neg A \wedge \neg B, \neg A, \neg B, A, B\}$

With these alternatives, Santorio's algorithm still predicts $\neg(A \wedge B)$ and $\neg A \vee \neg B$ to have the same truthmakers. Specifically, for (ii), the predicted truthmakers are $|\neg A|$ and $|\neg B|$, while for (iii), they are $|A \wedge \neg B|$, $|\neg A \wedge B|$, and $|\neg A \wedge \neg B|$.

[14] Proving this for Champollion and Bernard's (2024) system requires some mild assumptions: that A and B each have a verifier and their Cumulativity constraint—which they call 'natural'—generalised to the infinite case (for any nonempty $S = \{s_i\}_{i \in I}$ and $T = \{t_i\}_{i \in I}$, if s_i excludes t_i for all $i \in I$ then S's fusion excludes T's fusion).

this weekend implies permission to take Saturday off and permission to take Sunday off—the wrong result.

Moreover, truthmaker semantics of permission struggle with the Adam pair in (5). For plausibly, the embedded sentences 'Adam picks more than four pomegranates' and 'Adam picks five or more pomegranates' are not only necessarily equivalent but also *exactly* equivalent: they have the same truthmakers. The guiding idea behind truthmaker semantics is that a statement's truthmakers are 'wholly relevant' to its truth (Fine 2017b). Plausibly, the states that are wholly relevant to the truth of 'Adam picks more than four pomegranates' and 'Adam picks five or more pomegranates' are the states of Adam picking exactly n pomegranates, for $n = 5, 6, 7, \ldots$.[15] Given this, truthmaker semantics of permission fail to derive the contrast in (5).

There is also a more specific problem with Universal Realisability of Permission, concerning the coarseness of permission.

2.2 A Problem for Universal Realisability of Permission: Coarseness

Permission is *coarse* iff a statement can be permitted even if some ways for it to obtain are not.[16] Is permission coarse? Consider:

(9) a. There are two switches, A and B. Each switch can be either up or down. The rules specify that A must be up and that B may be in any position. Are the positions of the switches allowed to agree?

b. Alice and Bob are colleagues in Quebec. They know both English and French. A new law requires them to speak French at work, but they defy the law by talking to each other at work in English. Does the law permit Alice and Bob to talk to each other at work?

c. The client has instructed Alice to paint their house scarlet. Is Alice allowed to paint the house red?

d. Adam may pick at most five pomegranates. Is he allowed to pick more than four?

Intuitively, the answer to each is 'Yes'. For example, Alice and Bob are allowed to talk to each other at work, since they may do so in French, even though they may not do so in English.

It is plausible that the state of the switches being both down verifies that they agree, that the state of Alice and Bob talking to each other at work in English verifies that they talk to each other at work, that the state of Alice painting the house crimson (a shade of red other than scarlet) verifies that she paints it red, and that the state of Adam picking exactly six pomegranates verifies that he picks more than four (see Section 3.2 for some consequences of denying these assumptions). Given this, the judgements above are evidence against Universal Realisability of Permission and in favour of its existential variant:

Existential Realisability of Permission.
$\Diamond A$ is true iff some verifier of A is permitted.

We find further evidence for this principle in Alternative Eden:

[15] See Section 3.2 for some of the problems that result from dropping this assumption.

[16] The term 'coarseness' comes from Cariani (2013), who argues that *ought* is similarly coarse: *ought A* can be true even if some ways for A to hold are forbidden.

(10) Suppose God told Eve that she may eat every odd-numbered apple. Is Eve permitted to eat infinitely many of the apples?

Intuitively, she is. This is evidence that Existential Realisability is also correct for Fine's example. As Fine acknowledges,

> In the particular case of God's permitting Eve to eat infinitely many apples, we may obtain a perfectly good interpretation of the statement by taking it to be true just in case, for some infinite number of the apples, God has in fact granted Eve permission to eat those apples.
>
> (Fine 2014:333)

However, Fine's semantic clause does not respect his own observation. It instead predicts the statement 'Eve may eat infinitely many of the apples' to be true just in case, for *every* infinite collection of apples, God has granted Eve permission to eat those apples. Thus at the heart of Fine's paper lies an important tension: Fine acknowledges that existential realisability offers 'a perfectly good interpretation' of the permission statements in question, but his clauses enforce universal realisability.[17]

Universal Realisability of Permission seems hopelessly too strong. One might therefore consider adopting Existential Realisability of Permission (see e.g. Fine 2018:642). However, this also fails to predict the contrasts in (5) and in the De Morgan's Law example (7). Existential Realisability predicts $\Diamond(A \vee B)$ to be equivalent to $\Diamond A \vee \Diamond B$. It therefore predicts 'Adam may pick five or more pomegranates' to be true when he may pick at most five, and 'You may take off either Saturday or Sunday' to be true when you must work Saturday. In general, the problem is that Existential Realisability runs into the problem of free choice permission (Ross 1941, von Wright 1968, Kamp 1973, Merin 1992), which Fine (2018:641) explicitly wishes his theory to overcome.[18]

I conclude that both Universal and Existential Realisability of Permission face serious challenges. Before moving on, however, let us address two objections one may raise to try to rescue truthmaker semantics of permission from the problem of coarseness.

---

[17] One might reply that 'Eve may eat infinitely many of the apples' exhibits a scope ambiguity, with $\Diamond$ scoping either above or below *infinitely many*, as $\Diamond \bigvee_X p_X$ or $\bigvee_X \Diamond p_X$. It is, after all, well known that disjunction can take exceptionally high scope (Rooth and Partee 1982, Partee and Rooth 1983, Winter 2001, Schlenker 2006). One might then think that Fine's 'perfectly good interpretation' is simply due to the $\bigvee_X \Diamond p_X$ parse. This, however, fails to rescue universal realisability from the counterexamples in (9). For instance, there is no parse of 'The positions of the switches are allowed to agree' as 'There is a position P of the switches such that the switches are allowed to agree in P', but this is what the scope strategy would require. In addition, as we discuss in Section 3.1 below, the inference from 'Eve may eat infinitely many of the apples' to 'For every infinite collection of apples, Eve may eat it' is much weaker than the free choice inference from $\Diamond(A \vee B)$ to $\Diamond A \wedge \Diamond B$. But free choice exhibits the very same scope ambiguity: 'Eve may have an apple or a pear' may be parsed as either $\Diamond(A \vee B)$ or $\Diamond A \vee \Diamond B$ (to bring out the latter parse, consider 'Eve may have an apple or a pear, but I don't know which'; see Fusco 2019, Pinton 2021). The strategy of weakening by scope ambiguity would therefore incorrectly predict the same weakening for free choice.

[18] Fine explicitly appeals to free choice in support of his universal clause: 'suppose X is permissible. Then it is plausible (at least for those of us who have Ross-type intuitions!) that every action in compliance with X is permissible' (2018:640).

3 Objections and Replies

3.1 Objection 1: Universal Realisation Inferences

While the intuitive answer to the questions in (9) and (10) is 'Yes', it must be granted that asserting each permission statement outright will often trigger the respective inferences in (11).

(11) a. The positions of the switches are allowed to agree.
$\rightsquigarrow$ The switches are allowed to be both up and they are allowed to be both down.

b. Alice and Bob are allowed to talk to each other at work.
$\rightsquigarrow$ Alice and Bob are allowed to talk to each other at work in English and they are allowed to do so in French.

c. Alice is allowed to paint the house red.
$\rightsquigarrow$ Alice is allowed to paint the house any shade of red.

d. Adam may pick more than four pomegranates.
$\rightsquigarrow$ Adam may pick five, six, seven, … pomegranates.

e. Eve is permitted to eat infinitely many of the apples.
$\rightsquigarrow$ For every infinite collection of apples, Eve is permitted to eat every apple in the collection.

Each inference has the form $\Diamond A \rightsquigarrow (\Diamond B_0 \wedge \Diamond B_1 \wedge \cdots)$. In each case, it is natural to assume that for all i, every truthmaker for B_i is a truthmaker for A. Given this, the inferences in (11) are what we may call:

> *Universal realisation inferences.*
> The inference from $\Diamond A$ to: every truthmaker for A is permitted.

Universal Realisability of Permission derives these inferences as entailments. Is this not evidence for Universal Realisability of Permission?

I think not. Universal realisation inferences are best thought of as implicatures, derived by pragmatic reasoning. I show this below using two widely accepted tests for pragmatic inference: cancellability and question-sensitivity.

Test 1. Cancellability. Universal realisation inferences are cancellable. For instance, it is intuitively consistent to say, 'The positions of the switches are allowed to agree. More specifically, they are allowed to be both up, but they are not allowed to be both down'. If Universal Realisability of Permission were correct, the literal truth conditions of permission statements would entail their universal realisation inferences, so we would expect this to be equivalent to: 'Every way for the switches to agree is allowed. More specifically, they are allowed to be both up, but they are not allowed to be both down', which is inconsistent. Similarly, it is consistent to say, 'Eve may eat infinitely many of the apples. More specifically, she may only eat the odd-numbered apples' but inconsistent to say, 'For each infinite collection of apples, Eve may eat every apple in the collection. More specifically, she may only eat the odd-numbered apples'.

Test 2. Question-sensitivity. Pragmatic inferences are known to be highly sensitive to the question under discussion.[19] A classic example is the inference from *some* to *not all*. If Alice talked to all of the students, and Bob asks, 'Did Alice talk to some of the students?', it is natural to answer 'Yes'. If Bob had instead asked 'How many students did Alice talk to?', the answer 'Some of them' strongly suggests that she did not talk to all.

Universal realisation inferences are similarly question-sensitive. As we have seen, in response to the polar question 'Is A permitted?', as in (9) and (10), the natural answer is 'Yes'. In contrast, the inferences are reliably generated when the permission statements are not given in response to a Yes/No, *Is-this-permitted?* question, but in response to an open-ended *What-is-permitted?* question. In (11a), for example, if one is asked 'What positions of the switches are allowed?' or 'Tell me about the rules governing the switches', the answer 'The positions of the switches are allowed to agree' strongly implies that they may both be up and both be down. And if one is asked 'What apples may Eve eat?' or 'What may Eve do?', answering 'She may eat infinitely many of the apples' naturally suggests that for every infinite collection of the apples, Eve may eat them.

Universal realisation inferences are also reliably triggered on the prescriptive use of permission statements, which *create* permissions rather than merely describing those that already prevail.[20] One explanation for this is that permission statements on their prescriptive use answer the general *What-is-permitted?* question, which, as we have seen, readily triggers the universal realisation inferences.[21]

A paradigm case of the prescriptive use, if ever there was one, is God speaking to Eve. Universal realisation inferences may therefore explain the contrast between God's statements in Alternative Eden, (1) and (2). The difference is not a semantic difference in truth conditions, but a pragmatic difference in what inferences they trigger.

Free choice is not a universal realisation inference. It is worth distinguishing universal realisation inferences from the free choice inference from $\Diamond(A \vee B)$ to $\Diamond A$ and $\Diamond B$. Truthmaker semantics of permission assign them the same status. Free choice, however, is clearly more robust. It fails both of the tests for pragmatic inference above.

First, it is often observed that free choice inferences are not cancellable (Simons 2005, Barker 2010, Starr 2016, Aloni 2022, a.o.). Consider: 'Eve may have an apple or a pear. More specifically, she may have an apple but not a pear'. This sounds contradictory.[22]

[19] See e.g. Welker (1994), van Kuppevelt (1996), Breheny, Katsos and Williams (2006), Zondervan (2009), Degen and Tanenhaus (2015) and Ronai and Xiang (2021).

[20] For discussion of the difference between descriptive and prescriptive uses of permission statements, see Austin (1962), Kamp (1978), Lewis (1979), van Rooij (2000) and Kaufmann (2012), a.o.

[21] Note that universal realisation inferences do not *require* a prescriptive use. Imagine one who is informed about the relevant rules but lacks the authority to make them. When such a speaker asserts the statements in (11), we still tend to draw each inference, especially in response to a question about what is permitted in general (such as 'What positions of the switches are allowed?'). This shows that universal realisation inferences are distinct from the effects of deontic update, which have also been treated within truthmaker semantics (Yablo 2011, Fine 2018, Rothschild and Yablo 2023). In cases of deontic update, one with the relevant authority *changes* what is permitted, while universal realisation inferences may arise even for permission statements made by an independent observer with no power to change what is permitted.

[22] Further evidence that free choice inferences are not implicatures is that free choice inferences are processed faster than (Chemla

Second, free choice is not sensitive to the question under discussion in the way universal realisation inferences are. To illustrate, take 'Eve may have an apple or a pear'. The free choice inference that each option is permitted arises equally readily whether the statement answers the specific, *Is-this-permitted?* question, 'May Eve have an apple or a pear?' or the general, *What-is-permitted?* question, 'What may Eve eat?'.

Deriving Universal Realisation Inferences. To complete our account of universal realisation inferences, let us state how they may be derived. We do so by adapting the theory of Kratzer and Shimoyama (2002) to the truthmaker setting.[23] We take as our running example the inference from 'Alice may paint the house red' to 'She may paint it any shade of red', though our points apply generally.

Kratzer and Shimoyama begin by observing that statements are often interpreted exhaustively, as in (12).[24]

(12) Q: What colours is Alice allowed to paint the house?
A: She is allowed to paint it scarlet.
EXHAUSTIVE INFERENCE: No red shade other than scarlet is allowed.

They then consider a speaker who, when asked the question in (12), is deciding whether to give a more specific answer such as 'Scarlet' or a less specific answer such as 'Red'. In classical modal logic, the more specific answer is more informative than—i.e. entails—the less specific one. More informative utterances are in general better, along Gricean (1975) lines, at achieving the goal of cooperative conversation: to communicate information. If a speaker opts for a worse (less specific) utterance, principles of rational communication predict there to be a reason why they did not choose a better (more specific) one. Kratzer and Shimoyama point out that a prominent reason for not choosing the better one is that its exhaustive inference is false; in our example, that some red shade other than scarlet *is* allowed.

The same reasoning applies to any shade of red whatsoever. Answering 'Red' will convey that for every smaller set of red shades, a red shade outside of this set is permitted too. This derives precisely the inference we are after, that every shade of red is permitted.

We may adapt this reasoning to the truthmaker setting as follows. Let OK be the set of permitted states, and A a sentence whose set of truthmakers is T. We assume that for every more specific $T' \subsetneq T$, there is an alternative utterance B whose set of truthmakers is T'.[25] Given Existential Realisability of Permission, $\Diamond A$ expresses $\exists s \in T(s \in \text{OK})$. Uttering any more specific utterance B would have triggered the exhaustive inference $\neg\exists s \in T \setminus T'(s \in \text{OK})$. Since the speaker did not choose this, the listener may infer that its exhaustive inference is false: $\exists s \in T \setminus T'(s \in \text{OK})$. Since this reasoning applies to any more specific utterance whatsoever, for all $T' \subsetneq T$, $\exists s \in T \setminus T'(s \in OK)$. Hence $\forall s \in T(s \in \text{OK})$: every truthmaker for A is permitted—the desired result.[26]

and Bott 2014) and acquired before (Tieu et al. 2016) implicatures.

[23] One may plug in other accounts here, such as Iterated Best Response (Franke 2011).

[24] For discussion of exhaustive inferences, see Horn (1972), Groenendijk and Stokhof (1984), Spector (2003, 2007), Chierchia (2004), Schulz and van Rooij (2006), van Rooij and Schulz (2007), Fox (2007) and Franke (2011), among many others.

[25] This set of alternatives goes by the name *subdomain* alternatives in work on exhaustification; see e.g. Chierchia (2013).

[26] For the last step, pick any $s \in T$ and take $T' = T \setminus \{s\}$.

This account readily explains why universal realisation inferences are not derived in response to the question 'Is A permitted?'. For then the exhaustive inferences required by the derivation do not arise. For instance, if we replace the question in (12) with the *Is-this-permitted?* question, the exhaustive inference disappears.

(13) Q: Is Alice allowed to paint the house scarlet?
A: Yes, she is allowed to paint it scarlet.
~~EXHAUSTIVE INFERENCE: No red shade other than scarlet is allowed.~~

3.2 Objection 2: Denying the Truthmaking Assumptions

The coarseness of permission shows that $\Diamond A$ can be true even if some ways for A to obtain are forbidden. Coarseness is only a counterexample to Universal Realisability of Permission provided we take the 'ways' for a statement to obtain to be its truthmakers. This reflects the idea that truthmakers are fully determinate; for instance, that the truthmakers of 'The switches are in the same position' determine what position they are in. Specifically, the argument from coarseness against Universal Realisability of Permission assumed that each statement below has a truthmaker in common with the statement following it.

(14) a. The switches are in the same position.
The switches are both down.
b. Alice and Bob talk to each other at work.
Alice and Bob talk to each other at work in English.
c. Alice paints the house red.
Alice paints the house scarlet.
d. Adam picks more than four pomegranates.
Adam picks exactly six pomegranates.
e. Eve eats infinitely many of the apples.
Eve eats every apple in X, where X is an infinite set of apples containing an even-numbered apple.
f. $\neg(A \wedge B)$
$\neg A$

What would go wrong if we denied these assumptions? For instance, what if we denied that 'The switches are in the same position' shares a truthmaker with 'The switches are both down'? By parity of reasoning, the former would also share no truthmaker with 'The switches are both up'. Still, we would like 'The switches are in the same position' to have *some* truthmaker, to be able to apply truthmaker semantics for permission to this statement. The sole truthmakers for 'The switches are in the same position' would then be indeterminate states, verifying that the switches are in the same position but verifying neither that they are both up nor that they are both down.

At first sight, there is nothing particularly troubling about indeterminate states. We typically represent the world at a coarse level of granularity rather than in fundamental terms. In the truthmaker semantics literature, one regularly encounters phrases such as 'the state of the ball being red' without further refinement

as to its shade. And in the case of De Morgan's Law, Champollion and Bernard (2024) argue for the plausibility of what they call *emergent exclusion*, whereby an atomic state excludes the fusion of some atomic states without excluding any of the states individually. They give the example of three eggs and a basket with only space for two. Egg 1 being in the basket excludes that eggs 2 and 3 are both in the basket, but does not exclude that egg 2 is in the basket, nor that egg 3 is. In their system (a version of unilateral truthmaker semantics), an emergent excluder may verify $\neg(A \wedge B)$ without verifying either $\neg A$ or $\neg B$.

In general, let us associate with each indeterminate state a set of its *determinations*. For example, the determinations of the state of both switches being in the same position are the state of both being up and the state of both being down. The question then becomes how to determine when an indeterminate state is permitted. The following principle delivers the intuitive 'Yes' answers to our questions in (9).

(15) An indeterminate state is permitted iff at least one of its determinations is permitted.

In this way, indeterminate states allow us to reconcile Universal Realisability of Permission with coarseness.

On reflection, however, this move comes at a high cost. To reconcile Universal Realisability of Permission with coarseness, it is not enough to merely allow indeterminate states. We must take them to be the *sole* verifiers of the statements in question, allowing indeterminate and determinate states to coexist within one state space. This would do severe damage to the truthmaker semantics programme. Here are three problems.

First, we lose what is unique about truthmaker semantics: its exactness. Truthmaker semantics provides a distinctive notion of *exact* entailment, where A exactly entails B iff every verifier for A verifies B. If, say, 'the ball is red' were verified solely by an indeterminate state that does not settle its shade, we would lose the exact entailment from 'the ball is scarlet' to 'the ball is red'. This would have downstream consequences wherever exact entailment has been applied. The notion, for example, plays an essential role in truthmaker semantics of grounding (Fine 2012c). Plausible grounding claims would fail, such as (16).[27]

(16) a. The ball is red in virtue of being scarlet.
b. The switches are in the same position in virtue of being both down.
c. Adam picked more than four pomegranates in virtue of picking six pomegranates.
d. $\neg(A \wedge B)$ holds in virtue of $\neg A$; in other words, $\neg A$ grounds $\neg(A \wedge B)$

Second, the coexistence of determinate and indeterminate states within one state space is not compatible with Fine's (2011) treatment of the determinate/determinable distinction within a truthmaker setting. There Fine requires that the state space not contain both determinate and indeterminate states at once.[28]

Third, even predicting the *inexact* entailment from 'the ball is scarlet' to 'the ball is red' runs into trouble (where A inexactly entails B iff every truthmaker for A contains a truthmaker for B). This inexact entailment is motivated by the claim that 'scarlet' entails 'red' in virtue of the nature of the states involved; in this case, a mereological relationship. Thus any verifier of 'the ball is scarlet' (call it s_{scarlet}) should have as part the

[27] (16a) is endorsed by Rosen (2010), Audi (2012), Cameron (2016) and Bliss and Trogdon (2021), though see Wilson (2012) for a dissenting view.

[28] Specifically, see Fine (2011:168) on Directed Completeness.

indeterminate state verifying that the ball is red (s_{red}). However, this parthood relation has controversial consequences. Here are two.

1. Given that s_{red} is part of s_{scarlet}, the parthood relationship would have to be proper, to keep the two states distinct. But then standard principles of mereology—in particular, the widely endorsed principle of Weak Supplementation—imply the existence of a proper part of s_{scarlet} disjoint from s_{red}: a state that remains when one removes red from scarlet. However, it is a common refrain that the determinate–determinable relation lacks precisely this structure, and that such remainder states do not exist (Johnson 1921, Johansson 2000, Funkhouser 2006, Massin 2013, Wilson 2023).

2. On the truthmaker theory of partial content (see Fine 2017a:640–41), a statement is partly true iff it has a true part, where *A is part of B* iff every verifier of A is part of a verifier of B and every verifier of B contains a verifier of A. Given that s_{red} is part of s_{scarlet} and that 'the ball is red' is solely verified by s_{red}, 'the ball is red' is part of 'the ball is scarlet' in this sense. However, it then follows that for any ball that is red but not scarlet, it is partly true that it is scarlet. This is a bizarre consequence. Given a non-scarlet ball, the statement that it is scarlet, rather than partly true, seems entirely false.

In sum, while indeterminate states allow Universal Realisability of Permission to be compatible with coarseness, they raise a wide range of serious problems for applications of truthmaker semantics.

3.3 Interim Summary

Truthmaker semantics of permission and conditionals, at first sight, are well suited to account for Fine's Alternative Eden scenario. However, they face serious challenges. They do not account for failures of De Morgan's Law, and while modifications to the truthmaker clauses overcome the problem for conditionals, they continue to make incorrect predictions for permission, predicting $\Diamond\neg(A \wedge B)$ to entail $\Diamond\neg A$ and $\Diamond\neg B$. The coarseness of permission furthermore casts serious doubt on Universal Realisability of Permission, while Existential Realisability of Permission fails to account for free choice, for the contrast in (5), and for the De Morgan contrast in (7).

In response to these difficulties, it is worth exploring other accounts of our data. In what follows we consider a popular framework: alternative semantics.

4 Alternative Semantics of Modals and Conditionals

According to one line of thought, modals and conditionals are sensitive to *alternatives* (Aloni 2007, Alonso-Ovalle 2006, 2009, Ciardelli 2016, Cariani and Goldstein 2020). The general idea is this. We associate with each statement a set of propositions, called its 'alternatives', where a proposition is a set of worlds as in possible worlds semantics. For each sentence A, let $Alt(A)$ denote the set of A's alternatives. Formally, I assume an infinitary propositional language generated by $A ::= p \mid !A \mid \neg A \mid A \wedge B \wedge \cdots \mid A \vee B \vee \cdots \mid \Diamond A \mid A > C$. This language includes the flattening operator '!', familiar from alternative and inquisitive semantics, which takes any statement and returns one with a single alternative. We assume a valuation $|\cdot| : At \to \mathcal{P}(W)$ assigning to each atomic sentence the set of worlds where it is true. The clauses for the

connectives are as follows (Hamblin 1973, Kratzer and Shimoyama 2002, Alonso-Ovalle 2006).[29]

$$\begin{aligned} Alt(p) &= \{|p|\} \\ Alt(!A) &= \{\bigcup Alt(A)\} \\ Alt(\neg A) &= \{\overline{\bigcup Alt(A)}\} \\ Alt(A \wedge B \wedge \cdots) &= \{P \cap Q \cap \cdots : P \in Alt(A), Q \in Alt(B), \dots\} \\ Alt(A \vee B \vee \cdots) &= Alt(A) \cup Alt(B) \cup \cdots \end{aligned}$$

The underlying idea of these clauses is that overt disjunction introduces multiple alternatives, while the other connectives do not.

To derive an alternative semantics of modals and conditionals, we begin by assuming a basic theory expressed in terms of possible worlds. For modals, say, this could be truth in some accessible world; for conditionals, it could come from an underlying conditional relation $\Rightarrow$ between propositions (defined, say, in terms of similarity, premise semantics, interventions, and so on). To add alternatives, require that this basic meaning hold for *each* alternative.

Formally, let R be an accessibility relation between worlds and $\Rightarrow_w$ a world-dependent relation between propositions (e.g. $P \Rightarrow_w Q$ iff $f(P, w) \subseteq Q$, where f is a conditional selection function). Let $|C| = \bigcup Alt(C)$ be the set of worlds where C is true.

(17) *Alternative semantics for possibility.* (Aloni 2007)

a. $\Diamond A$ is true at w iff for all $P \in Alt(A)$, there is a $w' \in P$ such that wRw'

b. $Alt(\Diamond A) = \{\{w : \Diamond A \text{ is true at } w\}\}$

(18) *Alternative semantics for conditionals.* (Alonso-Ovalle 2006, 2009, Ciardelli 2016, Cariani and Goldstein 2020, a.o.)

a. $A > C$ is true at w iff for all $P \in Alt(A)$, $P \Rightarrow_w |C|$

b. $Alt(A > C) = \{\{w : A > C \text{ is true at } w\}\}$

Let us apply these clauses to Alternative Eden. To do so, we must decide what the alternatives to 'Eve eats infinitely many of $a_0, a_1, \dots$' should be. First, we might follow Fine (2014) in regimenting it as $\bigvee_{X \subseteq^\infty \{a_0, a_1, \dots\}} p_X$. This predicts its alternatives to be $\{|p_X| : X \subseteq^\infty \{a_0, a_1, \dots\}\}$, where p_X states that Eve eats all of X. Alternatively, we may use the flattening operator, taking 'Eve eats infinitely many of $a_0, a_1, \dots$' to express the flattened disjunction $!\bigvee_{X \subseteq^\infty \{a_0, a_1, \dots\}} p_X$, whose sole alternative is the set of worlds in which Eve eats infinitely many of the apples.

Neither choice gives us everything we want. The first assigns 'Eve may eat infinitely many of the apples' too strong a meaning. Like truthmaker semantics for permission, it predicts the statement to *entail* that for each infinite collection of apples, Eve may eat it. However, as we have seen, the statement is intuitively true when Eve, say, may eat any of the odd-numbered apples and none of the even-numbered apples. The second fails to distinguish between the conditionals in (3), since their antecedents express the same proposition, and on this option they each have this proposition as their sole alternative.

[29] For any set of worlds P, $\overline{P} = W \setminus P$ is P's complement.

5 A Positive Proposal

My proposal is to combine truthmaker semantics and alternative semantics. The semantic content of a statement is a set of propositions, as in alternative semantics, where each proposition is a set of states, as in truthmaker semantics.

This section shows how the resulting theory overcomes the challenges facing each framework in isolation. From truthmaker semantics, we take the ability to distinguish 'Eve eats infinitely many of a_0, a_1, a_2, ...' and 'Eve eats infinitely many of a_1, a_2, ...'. From alternative semantics, we take its treatment of disjunction. This will allow us to capture the contrast in the Adam pair in (5) and the violation of De Morgan's Law in (7). While truthmaker semantics does not distinguish them, alternative semantics does.[30]

The need to combine these two frameworks raises an important point: there are many roads to hyperintensionality. The Eve pair differs from the Adam and De Morgan pairs in how their hyperintensionality comes about. In the former it is due to a difference in their verifiers. In the latter it is due to a difference in their alternatives.

5.1 Alternative Truthmaker Semantics

Recall that according to alternative semantics, a statement's meaning is a set of propositions, where each proposition is a set of possible worlds. Our key change to alternative semantics is to replace sets of worlds with sets of truthmakers. To illustrate, let $|p|$ be the set of worlds where p is true, and $|p|^+$ the set of truthmakers of p. Thus where alternative semantics assigns $p \vee q$ the meaning $\{|p|, |q|\}$, we will assign it the meaning $\{|p|^+, |q|^+\}$. Naturally, we call the resulting theory *alternative truthmaker semantics*.

Formally, I adopt the language in Section 4 and assume the models of bilateral truthmaker semantics (Fine 2017b). In addition, we associate with each world w a set of states OK_w: the states that are permitted at w. Our models thus have the form $(S, \sqsubseteq, V^+, V^-, S^\Diamond, \text{OK})$, where

- S is a nonempty set of states;
- $\sqsubseteq$ is a partial order that is complete: every nonempty set of states T has a least upper bound, called its *fusion* and denoted $\bigsqcup T$;
- V^+ : $At \to \mathcal{P}(S)$ determines for each atomic sentence the set of its exact verifiers;
- V^- : $At \to \mathcal{P}(S)$ determines for each atomic sentence the set of its exact falsifiers;
- $S^\Diamond$ is a subset of states (intuitively, those that are possible) that is downward closed: if $s \in S^\Diamond$ and $t \sqsubseteq s$ then $t \in S^\Diamond$. Following Fine (2017b:561), we define a *world* w to be a maximal possible state (i.e. $w \in S^\Diamond$ and for all $s \in S$, if $s \sqcup w \in S^\Diamond$ then $s \sqsubseteq w$), and let W denote the set of worlds.
- OK : $W \to \mathcal{P}(S)$ assigns to each world a downward closed set of states: if $s \in \text{OK}_w$ and $t \sqsubseteq s$ then $t \in \text{OK}_w$.

Downward closure of OK_w states that permission for a composite state passes to its parts, guaranteeing the entailment from $\Diamond(A \wedge B)$ to $\Diamond A$.

[30] One may adopt other frameworks in place of alternative semantics here; most notably, inquisitive semantics (Ciardelli, Groenendijk and Roelofsen 2018). See Ciardelli, Roelofsen and Theiler (2017) for a comparison between the two frameworks. Nothing in what follows hinges on this choice.

Before giving the semantic clauses, one small issue must be addressed. Replacing sets of worlds with sets of truthmakers requires modifying the entry for conjunction. In standard alternative semantics, conjunction denotes pairwise intersection. In truthmaker semantics, conjunction is represented not by intersection but by pairwise fusion. We follow this treatment here. Given any (perhaps infinite) number of sets $P, Q, \dots$, let us abuse notation and denote their n-wise fusion as $P \sqcup Q \sqcup \dots = \{s \sqcup t \sqcup \dots : s \in P, t \in Q, \dots\}$.

The clauses for the connectives are then as follows, where $[\![A]\!]_M$ denotes the semantic value of a statement A in model M according to alternative truthmaker semantics (as before, I omit the subscript M for readability).

Semantic clauses in alternative truthmaker semantics.

$$
\begin{aligned}
[\![p]\!]^+ &= \{V^+(p)\} & [\![A \wedge B \wedge \dots]\!]^+ &= \{P \sqcup Q \sqcup \dots : P \in [\![A]\!]^+, Q \in [\![B]\!]^+, \dots\} \\
[\![p]\!]^- &= \{V^-(p)\} & [\![A \wedge B \wedge \dots]\!]^- &= [\![A]\!]^- \cup [\![B]\!]^- \cup \dots \\
[\![!A]\!]^+ &= \{\bigcup [\![A]\!]^+\} & [\![A \vee B \vee \dots]\!]^+ &= [\![A]\!]^+ \cup [\![B]\!]^+ \cup \dots \\
[\![!A]\!]^- &= \{\bigcup [\![A]\!]^-\} & [\![A \vee B \vee \dots]\!]^- &= \{P \sqcup Q \sqcup \dots : P \in [\![A]\!]^-, Q \in [\![B]\!]^-, \dots\} \\
[\![\neg A]\!]^+ &= \{\bigcup [\![A]\!]^-\} & & \\
[\![\neg A]\!]^- &= \{\bigcup [\![A]\!]^+\} & &
\end{aligned}
$$

We may call $[\![A]\!]^+$ the *positive alternatives* of A, and $[\![A]\!]^-$ its *negative alternatives*. We then take each statement's full semantic value to be the pair consisting of its positive and negative alternatives: $[\![A]\!] = ([\![A]\!]^+, [\![A]\!]^-)$.

From these clauses we may readily recover the familiar bilateral truthmaking clauses from Fine (2017b). For we have $\bigcup [\![A]\!]^+ = |A|^+$ and $\bigcup [\![A]\!]^- = |A|^-$ for any statement A, where $|A|^+$ denotes the set of its verifiers and $|A|^-$ the set of its falsifiers according to Fine's clauses.

Regarding truth, we define that A is *true* at a world w iff w contains a member of an alternative of A ($s \in P$ for some $s \sqsubseteq w$ and $P \in [\![A]\!]^+$), and define as usual that A *entails* B iff for every model M and world w of M, if A is true in w then B is true in w.

5.2 Making the Right Distinctions

The resulting theory is a 'two-storey' semantics, with two levels of hyperintensional content. On the lower level sit the truthmakers, on the higher level the alternatives. This is crucial for the theory to achieve its stated goal: distinguishing between the sentences in the Eve pair (4), the De Morgan pair (7), and the Adam pair (5). Concerning the first, 'Eve eats infinitely many of the apples $a_1, a_2, \dots$' and 'Eve eats infinitely many of the apples $a_0, a_1, a_2, \dots$' do not contain overt disjunction, so alternative semantics assigns them a single alternative. The present framework achieves this while maintaining Fine's disjunctive analysis of these sentences by flattening. That is, we parse them as $! \bigvee_{X \subseteq^\infty \{a_1, a_2, \dots\}} \bigwedge_{a \in X} p_a$ and $! \bigvee_{X \subseteq^\infty \{a_0, a_1, a_2, \dots\}} \bigwedge_{a \in X} p_a$, respectively. This returns:

(19) a. $[\![\text{Eve eats infinitely many of the apples } a_1, a_2, \dots]\!] = \{| \bigvee_{X \subseteq^\infty \{a_1, a_2, \dots\}} \bigwedge_{a \in X} p_a |^+\}$

b. $[\![\text{Eve eats infinitely many of the apples } a_0, a_1, a_2, \dots]\!] = \{| \bigvee_{X \subseteq^\infty \{a_0, a_1, a_2, \dots\}} \bigwedge_{a \in X} p_a |^+\}$

Each sentence has one alternative—the set of its truthmakers—and since they have different truthmakers, they correspondingly have different alternatives too.

Alternative truthmaker semantics accounts for the contrast in (7) by violating De Morgan's Law. $\neg(A \wedge B)$ and $\neg A \vee \neg B$ have the same truthmakers but different alternatives.

$$|\neg(A \wedge B)|^+ = |\neg A \vee \neg B|^+ \qquad\qquad [\![\neg(A \wedge B)]\!]^+ \neq [\![\neg A \vee \neg B]\!]^+$$

This is because negation flattens alternatives while disjunction introduces them: $[\![\neg(p \wedge q)]\!]^+ = \{|p|^- \cup |q|^-\}$ has one alternative, while $[\![\neg p \vee \neg q]\!]^+ = \{|p|^-, |q|^-\}$ has two.

Turning to the Adam pair, assuming that 'Adam picks more than four pomegranates' and 'Adam picks five or more pomegranates' have the same truthmakers, they are instead distinguished by their alternatives. Within the present language, let us regiment 'Adam picks exactly n pomegranates' and 'Adam picks more than n pomegranates' as atomic sentences, denoted p_n and $p_{>n}$, respectively. Since $p_{>4}$ is not a disjunction, it has a single alternative, while $p_5 \vee p_{>5}$ is a disjunction, with two alternatives.

$$|p_{>4}|^+ = |p_5 \vee p_{>5}|^+ \qquad\qquad [\![p_{>4}]\!]^+ \neq [\![p_5 \vee p_{>5}]\!]^+$$

5.3 Adding Permission

Now that we have a semantic framework that determines the semantic content of each statement, let us state what it means for a statement to be permitted. Our starting point is Aloni's (2007) alternative semantics of modals in (17a): $\Diamond A$ is true iff each of A's alternatives contains a deontically ideal world. To transpose Aloni's clause into alternative *truthmaker* semantics, I will modify it in two ways. The first change is to replace the possible worlds clause for permission (truth in some deontically ideal world) with its truthmaker analogue, that some verifier is permitted. The second change is to assume, following Goldstein (2019), that alternatives are subject to a homogeneity presupposition to the effect that the alternatives have the same status: either every alternative contains a permitted state or none does. This all-or-none condition will ensure *Dual Prohibition*, the inference from $\neg\Diamond(A \vee B)$ to $\neg\Diamond A$ and $\neg\Diamond B$; for instance, that 'Alice may not have ice cream or cake' implies that she may not have ice cream *and* that she may not have cake.

Given these two changes, I propose the following truth conditions for permission statements.

(20) *Alternative truthmaker semantics of permission.*

a. $\Diamond A$ is defined at w only if $\forall P, Q \in [\![A]\!]^+ : (\exists s \in P : s \in \mathrm{OK}_w)$ iff $(\exists s \in Q : s \in \mathrm{OK}_w)$;

b. if defined, $\Diamond A$ is true at w iff $\forall P \in [\![A]\!]^+ \; \exists s \in P : s \in \mathrm{OK}_w$.

This says that a permission statement $\Diamond A$ is defined only if all or none of A's alternatives contain a permitted state; and if defined, is true iff all of them do.[31,32]

[31] Given this homogeneity presupposition and the fact that every sentence has some alternative (ensured by the semantic clauses), appeal to universal quantification in (20b) is inessential: we may equivalently take $\Diamond A$, if defined, to be true iff *some* alternative of A contains a permitted state.

[32] The reader may wonder what this theory has to say about obligation. While we are squarely concerned with permission here, given Duality one may extract a clause for obligation by analysing $\Box A$ as $\neg\Diamond\neg A$. Since negations have a single alternative, the homogeneity presupposition becomes inert, and the resulting clause is that $\Box A$ is true at w iff $\forall s \in |A|^- : s \notin \mathrm{OK}_w$, i.e. A is

This clause has the desired results. It guarantees free choice, that $\Diamond(A \vee B)$ implies $\Diamond A$ and $\Diamond B$. It captures the failure of De Morgan's Law and the contrast in the Adam pair. Given that Adam may pick at most five pomegranates, 'Adam may pick more than four pomegranates' is true: $[\![p_{>4}]\!]^+ = \{|p_{>4}|^+\}$ and some exact verifier of $p_{>4}$ is permitted; namely, picking exactly five pomegranates. But 'Adam may pick five or more pomegranates' is not true—rather, it is undefined—since $[\![p_5 \vee p_{>5}]\!]^+ = \{|p_5|^+, |p_{>5}|^+\}$, but the alternatives differ in status: picking exactly five pomegranates is permitted, while picking more than five is not.

Turning to the failure of De Morgan's Law in (7), where p denotes working Saturday and q working Sunday, $[\![\neg(p \wedge q)]\!]^+ = \{|p|^- \cup |q|^-\}$ while $[\![\neg p \vee \neg q]\!]^+ = \{|p|^-, |q|^-\}$. Hence one has permission to not work both days this weekend, $\Diamond\neg(p \wedge q)$, iff some falsifier of 'work Saturday' or 'work Sunday' is permitted. This is compatible with a requirement to work Saturday. But one has permission to take Saturday or Sunday off, $\Diamond(\neg p \vee \neg q)$, iff some falsifier of 'work Saturday' is permitted *and* some falsifier of 'work Sunday' is permitted. This is not compatible with a requirement to work Saturday.

5.4 Adding Conditionals

Following alternative semantics of conditionals, I assume a basic, world-dependent conditional relation $\Rightarrow_w$. Instead of relating the propositions of possible worlds semantics (sets of worlds), it relates those of truthmaker semantics (sets of states). This may be analysed in terms of existing state-based theories of conditionals, such as Fine's (2012b) truthmaker semantics of counterfactuals or McHugh's (2023) aboutness theory of counterfactuals. Following Fine's approach, for any sets of states P and Q we take $P \Rightarrow_w Q$ to hold iff for every state s in P and possible outcome t of s at w, t contains a state in Q ($\forall s \in P\, \forall t(s \rightarrow_w t \supset \exists u \in Q : u \sqsubseteq t)$, where $\rightarrow_w$ is Fine's possible outcome relation). Following McHugh's, we take $P \Rightarrow_w Q$ iff every world that (i) results from varying the states that P is about at w and playing the laws forward and (ii) contains a state in P, contains a state in Q (see McHugh 2022, 2023:125–26 for a formalisation).

Given one's preferred analysis of $\Rightarrow_w$, we lift this basic conditional relation to the level of alternatives following the recipe laid out by Alonso-Ovalle (2009), Ciardelli (2016) and Santorio (2018): take conditionals to quantify universally over their antecedents' alternatives. To this condition we add homogeneous quantification over alternatives, following Cariani and Goldstein (2020) and mirroring the clause for permission in (20).

(21) *Alternative truthmaker semantics of conditionals.*

a. $A > C$ is defined at w only if $\forall P, Q \in [\![A]\!]^+ : P \Rightarrow_w |C|^+$ iff $Q \Rightarrow_w |C|^+$;

b. if defined, $A > C$ is true at w iff $\forall P \in [\![A]\!]^+ : P \Rightarrow_w |C|^+$.

The homogeneity clause in (21a) is designed to account for an observation due to Santorio (2018:524), that $\neg((A \vee B) > C)$ implies both $\neg(A > C)$ and $\neg(B > C)$, illustrated in (22).

(22) No girl would be happy if she failed algebra or geography.

obligatory at w iff no falsifier for A is permitted. To predict the inference from $\Box(A \vee B)$ to $\Diamond A \wedge \Diamond B$, we may follow Aloni (2007:86) and Fine (2018:641) in taking obligation to express $\Box^\Diamond$, with $\Box^\Diamond A$ defined as $\Box A \wedge \Diamond A$.

a. $\rightsquigarrow$ No girl would be happy if she failed algebra.

b. $\rightsquigarrow$ No girl would be happy if she failed geography.

Our clause for the conditional accounts for the conditional Eve pair in (3). Defining $\Rightarrow_w$ over sets of truthmakers allows Conditional Substitution to fail, even for sentences like those in the Eve pair with a single alternative.[33]

It also accounts for failures of De Morgan's Law. $(\overline{A} \vee \overline{B}) > \textsc{Off}$ is true iff $(\overline{A} > \textsc{Off}) \wedge (\overline{B} > \textsc{Off})$ is. On the natural implementation of $\Rightarrow_w$ within Fine's and McHugh's theories, each conjunct comes out true, predicting (8a) as a whole to be true. However, (8b), $\neg(A \wedge B) > \textsc{Off}$, is true iff $|A \wedge B|^- \Rightarrow_w |\textsc{Off}|^+$. Its status therefore depends on the underlying conditional $\Rightarrow_w$. Both Fine's and McHugh's theories also have the resources to predict its falsity, and therefore to account for the contrast in (8). For Fine, we achieve this by adopting the *inclusive* falsification clause for conjunction (while retaining the non-inclusive clause for disjunction), according to which a state falsifies $A \wedge B$ iff it falsifies A, B, or is the fusion of a falsifier of A with a falsifier of B.[34] For then the state of both switches being down verifies that they are not both up, predicting $\neg(A \wedge B) > \textsc{Off}$ to entail $(\neg A \wedge \neg B) > \textsc{Off}$, which is false. On McHugh's aboutness approach, we achieve this because $\neg(A \wedge B)$ is about the states of both switches, and varying them simultaneously allows both to be down (see McHugh 2023:143).

5.5 Verification and Falsification Clauses

Our clauses above provide the truth conditions for permission statements and conditionals. They do not, however, tell us which states verify or falsify such statements. This would be required to give recursive clauses in full generality, to determine exact entailment (where one statement exactly entails another iff every verifier for the first verifies the second), as well as to preserve the familiar idea that a statement is true (respectively, false) at a world iff the world contains an exact verifier (falsifier) of the statement. Let us do this for permission first, and then conditionals.

We follow Fine (2018:644) in introducing designated states representing the state of the particular code of conduct prevailing. Formally, for any set of states $P \subseteq S$, we introduce a state $s_P^\Diamond$—the state of P being all and only the permitted states—subject to the requirement that $s_P^\Diamond$ obtain at (i.e. be part of) a world w iff $P = \text{OK}_w$.[35] We then define that such a state $s_P^\Diamond$ verifies $\Diamond A$ iff $\Diamond A$ is true when P is the set of permitted states, and falsifies $\Diamond A$ iff $\Diamond A$ is false when P is the set of permitted states. Since permission statements are

[33] Where P_0 is the set of verifiers of 'Eve eats infinitely many of $a_0, a_1, a_2, \ldots$', P_1 the set of verifiers of 'Eve eats infinitely many of $a_1, a_2, \ldots$', and Q the set of verifiers of 'Eve is OK', the reader may readily verify that Fine's and McHugh's theories predict $P_0 \Rightarrow_w Q$ to be false and $P_1 \Rightarrow_w Q$ to be true; for details, see Fine (2012b) and McHugh (2023:157).

[34] This follows the strategy from Bernard and Champollion (2018), Champollion and Bernard (2024), Li (2023) and De Rizzo (2025), discussed in Section 2.1. In alternative truthmaker semantics, the inclusive falsification clause for conjunction is $[\![\bigwedge_{i\in I} A_i]\!]^- = \bigcup_{\emptyset\neq J\subseteq I} \bigsqcup_{j\in J} [\![A_j]\!]^-$ (and the inclusive verification clause for disjunction is $[\![\bigvee_{i\in I} A_i]\!]^+ = \bigcup_{\emptyset\neq J\subseteq I} \bigsqcup_{j\in J} [\![A_j]\!]^+$), where $\bigsqcup$ denotes the n-wise fusion operation from Section 5.1.

[35] I assume this operation to be injective: $s_P^\Diamond = s_{P'}^\Diamond$ iff $P = P'$, for any $P, P' \subseteq S$. Note that the states $s_P^\Diamond$ must not be in S, but rather in an enlarged state space $S' \supsetneq S$. Otherwise we would run into a cardinality paradox: each $s_P^\Diamond$ corresponds to a subset of S, so by Cantor's theorem, there are strictly more states $s_P^\Diamond$ than states in S. This new set S' will then play the role of our state space.

not disjunctive, they have a single alternative.

$$[\![\Diamond A]\!]^+ = \{\{s_P^\Diamond : \forall Q \in [\![A]\!]^+ \; \exists s \in Q, s \in P\}\}$$
$$[\![\Diamond A]\!]^- = \{\{s_P^\Diamond : \forall Q \in [\![A]\!]^+ \; \neg\exists s \in Q, s \in P\}\}$$

These automatically incorporate our homogeneity assumption. Statements where one alternative is permitted and another is not—as in 'You may post the letter or burn down the post office'—will have neither a verifier nor a falsifier. And on the standard assumption that a statement is true (respectively, false) at a world iff the world contains an exact verifier (falsifier) of it, the statement will be neither true nor false at worlds containing neither a verifier nor a falsifier of it.

We make an analogous move for conditionals. For any binary relation R between sets of states, introduce the state $s_R^{\Rrightarrow}$, thought of as the state of R being the obtaining conditional relation, subject to the requirement that $s_R^{\Rrightarrow}$ obtain at a world w iff $R = \Rightarrow_w$.[36]

$$[\![A > C]\!]^+ = \{\{s_R^{\Rrightarrow} : \forall P \in [\![A]\!]^+, R(P, |C|^+)\}\}$$
$$[\![A > C]\!]^- = \{\{s_R^{\Rrightarrow} : \forall P \in [\![A]\!]^+, \neg R(P, |C|^+)\}\}$$

These also incorporate our homogeneity assumption. For example, if $p > r$ is true but $q > r$ is false at w (i.e. $|p|^+ \Rightarrow_w |r|^+$ but $|q|^+ \not\Rightarrow_w |r|^+$), then w will contain neither a verifier nor a falsifier of $(p \vee q) > r$.

5.6 Comparison with previous approaches

Alternative truthmaker semantics accounts for all the data so far discussed. This is shown in Table 1, which provides an overview of the theories we have discussed and their predictions.

6 Conclusion

Modals and conditional antecedents represent a rich and influential case study in the limits of possible worlds semantics. Extant approaches—in particular, alternative and truthmaker semantics—have failed to properly attend to the different ways in which the hyperintensional behaviour of permission and conditionals comes about. Some failures of substitution of necessary equivalents result from differences in truthmakers, others from differences in alternatives. Alternative semantics accounts for the behaviour of disjunction, predicting the contrast in the Adam pair and the failure of De Morgan's Law. Truthmaker semantics accounts for the behaviour of *infinitely many*, giving correct predictions for Fine's Alternative Eden scenario. While each framework in isolation faces serious challenges, their combination overcomes them, allowing us to take the best of both worlds. What emerges is a two-storey semantics, with two levels of hyperintensional content, each making its own essential contribution. It is time to leave our separate bungalows and upgrade to a two-storey house.

[36] As before, I assume injectivity, $s_R^{\Rrightarrow} = s_{R'}^{\Rrightarrow}$ iff $R = R'$, for all $R, R' \subseteq \mathcal{P}(S) \times \mathcal{P}(S)$, and that any $s_R^{\Rrightarrow}$ is not in S but in the larger state space S'.

Table 1: Predictions of the theories discussed.

Theory	Coarseness	Infinitely many (>)	Free choice	De Morgan (◇)	De Morgan (>)	Proposed by
Possible worlds semantics	✓	✗	✗	✓	✗	◇: Hanson 1965, Kratzer 1977; >: Stalnaker 1968, Lewis 1973
Truthmaker semantics + ∀-realisability	✗	✓	✓	✗	✗	◇: Fine 2014, 2018, Anglberger, Korbmacher and Faroldi 2016. >: Fine 2012a,b, Briggs 2012, Santorio 2018, Rosella and Sprenger 2024, Brast-McKie 2025
Truthmaker, ∀-realisability + De Morgan-breaking clauses	✗	✓	✓	✗	✓	De Morgan-breaking clauses: Bernard and Champollion 2018, Champollion and Bernard 2024, Li 2023, De Rizzo 2025
Truthmaker, ∃-realisability of ◇ + ∀-realisability of > + De Morgan-breaking clauses	✓	✓	✗	✓	✓	◇: Section 2.2 (cf. Fine 2018:642). >: as in the two rows above
Alternatives, unflattened	✗	✓	✓	✓	✓	◇: Aloni 2007, Goldstein 2019. >: Alonso-Ovalle 2006, 2009, Ciardelli 2016, Cariani and Goldstein 2020. Regimentation: Fine 2014, with unflattened *infinitely many*: $\bigvee_X p_X$
Alternatives, flattened	✓	✗	✓	✓	✓	As above, with flattened *infinitely many*: $!\bigvee_X p_X$
Alternative truthmaker semantics	✓	✓	✓	✓	✓	This paper. ⇒ from Fine 2012b or McHugh 2022, 2023. Homogeneity from Goldstein 2019, Cariani and Goldstein 2020

Coarseness: 'Eve may eat infinitely many apples' is literally true in Alternative Eden, and 'The switches are allowed to be in the same position' is literally true when they may both be up but may not both be down. *Infinitely many* (>): the conditionals in (3) differ in truth value. *Free choice*: $\Diamond(A \vee B)$ entails $\Diamond A \wedge \Diamond B$. *De Morgan* (◇): $\Diamond\neg(A \wedge B)$ is compatible with $\neg\Diamond\neg A$, as in (7a). *De Morgan* (>): (8a) is true and (8b) is not.

References

Aloni, Maria (2007). Free choice, modals, and imperatives. *Natural Language Semantics* 15.1, pp. 65–94. DOI: 10.1007/s11050-007-9010-2.

— (2018). Disjunctions in state-based semantics. *ILLC preprint*.

— (2022). Logic and conversation: The case of free choice. *Semantics and Pragmatics* 15.5. DOI: 10.3765/sp.15.5.

Alonso-Ovalle, Luis (2006). Disjunction in alternative semantics. PhD thesis. University of Massachusetts Amherst. URL: http://people.linguistics.mcgill.ca/~luis.alonso-ovalle/papers/alonso-ovalle-diss.pdf.

— (2009). Counterfactuals, correlatives, and disjunction. *Linguistics and Philosophy* 32.2, pp. 207–244. DOI: 10.1007/s10988-009-9059-0.

Anglberger, Albert J. J., Norbert Gratzl and Olivier Roy (2015). Obligation, Free Choice, and the Logic of Weakest Permissions. *The Review of Symbolic Logic* 8.4, pp. 807–827. DOI: 10.1017/S1755020315000209.

Anglberger, Albert J. J., Johannes Korbmacher and Federico L. G. Faroldi (2016). An Exact Truthmaker Semantics for Permission and Obligation. *Deontic Logic and Normative Systems: 13th International Con-*

ference, DEON 2016, Bayreuth, Germany, July 18–21, 2016. Ed. by Olivier Roy, Allard Tamminga and Malte Willer. London: College Publications, pp. 16–31.

Asher, Nicholas and Daniel Bonevac (2005). Free choice permission is strong permission. *Synthese* 145.3, pp. 303–323. DOI: 10.1007/s11229-005-6196-z.

Audi, Paul (2012). Grounding: toward a theory of the *in-virtue-of* relation. *The Journal of Philosophy* 109.12, pp. 685–711. DOI: 10.5840/jphil20121091232.

Austin, J. L. (1962). *How to Do Things with Words*. Oxford: Clarendon Press.

Barker, Chris (2010). Free choice permission as resource-sensitive reasoning. *Semantics and Pragmatics* 3, pp. 1–38. DOI: 10.3765/sp.3.10.

Bernard, Timothée and Lucas Champollion (2018). Negative events in compositional semantics. *Semantics and Linguistic Theory*. Vol. 28, pp. 512–532. DOI: 10.3765/salt.v28i0.4429.

Bliss, Ricki and Kelly Trogdon (2021). Metaphysical grounding. *The Stanford Encyclopedia of Philosophy*. Ed. by Edward N. Zalta. Winter 2021. Metaphysics Research Lab, Stanford University. URL: https://plato.stanford.edu/archives/win2021/entries/grounding.

Brast-McKie, Benjamin (2025). Counterfactual Worlds. *Journal of Philosophical Logic* 54.3, pp. 533–574. DOI: 10.1007/s10992-025-09793-8.

Breheny, Richard, Napoleon Katsos and John Williams (2006). Are Generalised Scalar Implicatures Generated by Default? An On-line Investigation into the Role of Context in Generating Pragmatic Inferences. *Cognition* 100.3, pp. 434–463. DOI: 10.1016/j.cognition.2005.07.003.

Briggs, Ray (2012). Interventionist counterfactuals. *Philosophical Studies* 160.1, pp. 139–166. DOI: 10.1007/s11098-012-9908-5.

Cameron, Ross P. (2016). Do we need grounding? *Inquiry* 59.4, pp. 382–397. DOI: 10.1080/0020174X.2015.1128848.

Cariani, Fabrizio (2013). 'Ought' and resolution semantics. *Noûs* 47.3, pp. 534–558. DOI: 10.1111/j.1468-0068.2011.00839.x.

Cariani, Fabrizio and Simon Goldstein (2020). Conditional Heresies. *Philosophy and Phenomenological Research* 101.2, pp. 251–282. DOI: 10.1111/phpr.12565.

Champollion, Lucas and Timothée Bernard (2024). Negation and modality in unilateral truthmaker semantics. *Linguistics and Philosophy* 47.4, pp. 517–555. DOI: 10.1007/s10988-023-09407-z.

Chemla, Emmanuel and Lewis Bott (2014). Processing inferences at the semantics/pragmatics frontier: Disjunctions and free choice. *Cognition* 130.3, pp. 380–396. DOI: 10.1016/j.cognition.2013.11.013.

Chierchia, Gennaro (2004). Scalar Implicatures, Polarity Phenomena and the Syntax/Pragmatics Interface. *Structures and Beyond*. Oxford University Press, pp. 39–103.

— (2013). *Logic in Grammar*. Oxford University Press. DOI: 10.1093/acprof:oso/9780199697977.001.0001.

Chrisman, Matthew (2015). *The Meaning of 'Ought': Beyond Descriptivism and Expressivism in Metaethics*. Oxford Moral Theory. New York: Oxford University Press. DOI: 10.1093/acprof:oso/9780199363001.001.0001.

Ciardelli, Ivano (2016). Lifting conditionals to inquisitive semantics. *Semantics and Linguistic Theory*. Vol. 26, pp. 732–752. DOI: 10.3765/salt.v26i0.3811.

Ciardelli, Ivano, Jeroen Groenendijk and Floris Roelofsen (2018). *Inquisitive Semantics*. Oxford Surveys in Semantics and Pragmatics. Oxford University Press. DOI: 10.1093/oso/9780198814788.001.0001.

Ciardelli, Ivano, Floris Roelofsen and Nadine Theiler (2017). Composing alternatives. *Linguistics and Philosophy* 40.1, pp. 1–36. DOI: 10.1007/s10988-016-9195-2.

Ciardelli, Ivano, Linmin Zhang and Lucas Champollion (2018). Two switches in the theory of counterfactuals: A study of truth conditionality and minimal change. *Linguistics and Philosophy* 41.6, pp. 577–621. DOI: 10.1007/s10988-018-9232-4.

De Rizzo, Julio (2025). Simpler Representational Ground. *Erkenntnis* 90.8, pp. 3809–3832. DOI: 10.1007/s10670-024-00875-8.

Degen, Judith and Michael K. Tanenhaus (2015). Processing Scalar Implicature: A Constraint-Based Approach. *Cognitive Science* 39.4, pp. 667–710. DOI: 10.1111/cogs.12171.

Fine, Kit (2011). An Abstract Characterization of the Determinate/Determinable Distinction. *Philosophical Perspectives* 25.1, pp. 161–187. DOI: 10.1111/j.1520-8583.2011.00224.x.

— (2012a). A difficulty for the possible worlds analysis of counterfactuals. *Synthese* 189.1, pp. 29–57. DOI: 10.1007/s11229-012-0094-y.

— (2012b). Counterfactuals without possible worlds. *Journal of Philosophy* 109.3, pp. 221–246. DOI: 10.5840/jphil201210938.

— (2012c). Guide to ground. *Metaphysical Grounding: Understanding the Structure of Reality*. Ed. by Fabrice Correia and Benjamin Schnieder. Cambridge University Press, pp. 37–80. DOI: 10.1017/CBO9781139149136.002.

— (2014). Permission and possible worlds. *dialectica* 68.3, pp. 317–336. DOI: 10.1111/1746-8361.12068.

— (2017a). A theory of truthmaker content I: Conjunction, disjunction and negation. *Journal of Philosophical Logic* 46.6, pp. 625–674. DOI: 10.1007/s10992-016-9413-y.

— (2017b). Truthmaker Semantics. *A Companion to the Philosophy of Language*. Ed. by Bob Hale, Crispin Wright and Alexander Miller. Wiley-Blackwell. Chap. 22, pp. 556–577. DOI: 10.1002/9781118972090.ch22.

— (2018). Compliance and Command II, Imperatives and Deontics. *The Review of Symbolic Logic* 11.4, pp. 634–664. DOI: 10.1017/S1755020318000059.

Fox, Danny (2007). Free choice and the theory of scalar implicatures. *Presupposition and Implicature in Compositional Semantics*. Ed. by Uli Sauerland and Penka Stateva. Palgrave, pp. 71–120. DOI: 10.1057/9780230210752_4.

Franke, Michael (2011). Quantity implicatures, exhaustive interpretation, and rational conversation. *Semantics and Pragmatics* 4.1, pp. 1–82. DOI: 10.3765/sp.4.1.

Funkhouser, Eric (2006). The Determinable–Determinate Relation. *Noûs* 40.3, pp. 548–569. DOI: 10.1111/j.1468-0068.2006.00623.x.

Fusco, Melissa (2015). Deontic Modality and the Semantics of Choice. *Philosophers' Imprint* 15.28, pp. 1–27. URL: http://hdl.handle.net/2027/spo.3521354.0015.028.

— (2019). Sluicing on free choice. *Semantics and Pragmatics* 12.20. DOI: 10.3765/sp.12.20.

Goldstein, Simon (2019). Free choice and homogeneity. *Semantics and Pragmatics* 12.23, pp. 1–47. DOI: 10.3765/sp.12.23.

Goodsell, Zachary (2022). Tossing Morgenbesser's Coin. *Analysis* 82.2, pp. 214–221. DOI: 10.1093/analys/anab091.

Grice, Herbert Paul (1975). Logic and conversation. *Speech acts*. Brill, pp. 41–58.

Groenendijk, Jeroen and Martin Stokhof (1984). Studies on the semantics of questions and the pragmatics of answers. PhD thesis. University of Amsterdam. URL: http://hdl.handle.net/11245/1.392528.

Hamblin, Charles L. (1973). Questions in Montague English. *Foundations of Language* 10.1, pp. 41–53.

Hanson, William H. (1965). Semantics for deontic logic. *Logique et Analyse* 8.31, pp. 177–190.

Hansson, Sven Ove (2013). The Varieties of Permissions. *Handbook of Deontic Logic and Normative Systems*. Ed. by Dov Gabbay et al. Vol. 1. London: College Publications.

Horn, Laurence (1972). On The Semantic Properties of Logical Operators in English. PhD thesis. University of California Los Angeles.

Johansson, Ingvar (2000). Determinables as Universals. *The Monist* 83.1, pp. 101–121. DOI: 10.5840/monist20008312.

Johnson, W. E. (1921). *Logic, Part I*. Cambridge: Cambridge University Press.

Kamp, Hans (1973). Free choice permission. *Proceedings of the Aristotelian Society* 74.1, pp. 57–74. DOI: 10.1093/aristotelian/74.1.57.

— (1978). Semantics versus pragmatics. *Formal Semantics and Pragmatics for Natural Languages*. Ed. by Franz Guenthner and Siegfried J. Schmidt. Vol. 4. Synthese Language Library. Dordrecht: Reidel, pp. 255–287.

Katzir, Roni (2007). Structurally-defined alternatives. *Linguistics and Philosophy* 30.6, pp. 669–690. DOI: 10.1007/s10988-008-9029-y.

Kaufmann, Magdalena (2012). *Interpreting Imperatives*. Studies in Linguistics and Philosophy 88. Dordrecht: Springer. DOI: 10.1007/978-94-007-2269-9.

Kratzer, Angelika (1977). What 'must' and 'can' must and can mean. *Linguistics and Philosophy* 1.3, pp. 337–355. DOI: 10.1007/BF00353453.

Kratzer, Angelika and Junko Shimoyama (2002). Indeterminate pronouns: The view from Japanese. *The Proceedings of the Third Tokyo Conference on Psycholinguistics*. Ed. by Y. Otsu, pp. 1–25. URL: https://people.umass.edu/partee/RGGU_2004/Indeterminate%20Pronouns.pdf.

van Kuppevelt, Jan (1996). Inferring from topics: Scalar implicatures as topic-dependent inferences. *Linguistics and Philosophy* 19, pp. 393–443. DOI: 10.1007/BF00630897.

Lewis, David (1970). General Semantics. *Synthese* 22, pp. 18–67. DOI: 10.1007/BF00413598.

— (1973). *Counterfactuals*. Oxford: Blackwell.

Lewis, David (1979). A problem about permission. *Essays in Honour of Jaakko Hintikka: On the Occasion of His Fiftieth Birthday*. Ed. by Esa Saarinen et al. D. Reidel, pp. 163–175. DOI: 10.1007/978-94-009-9860-5_11.

— (1986). *On the plurality of worlds*. Oxford: Blackwell.

Li, Mingyuan (2023). Truthmaker and Negation: Investigations of Nonbeing, Negative Events, and Counterfactual Conditionals. PhD thesis. University of Illinois at Urbana-Champaign. URL: https://hdl.handle.net/2142/122098.

Lokhorst, Gert-Jan C. (1997). *Deontic Linear Logic with Petri Net Semantics*. Technical Report. Rotterdam. URL: https://gjclokhorst.nl/deopetri.pdf.

Marty, Paul et al. (2021). Negative Free Choice. *Semantics and Pragmatics* 14.13, pp. 1–54. DOI: 10.3765/sp.14.13.

Massin, Olivier (2013). Determinables and Brute Similarities. *Johanssonian Investigations. Essays in Honour of Ingvar Johansson on His Seventieth Birthday*. Ed. by Christer Svennerlind, Jan Almäng and Rögnvaldur Ingthorsson. Heusenstamm: Ontos Verlag, pp. 388–420. DOI: 10.1515/9783110322507.388.

McHugh, Dean (2022). Aboutness and modality. *Proceedings of the 23rd Amsterdam Colloquium*, pp. 194–206. DOI: 10.21942/uva.21739718.

— (2023). Causation and Modality: Models and Meanings. PhD thesis. University of Amsterdam. URL: https://eprints.illc.uva.nl/id/eprint/2243.

Merin, Arthur (1992). Permission Sentences Stand in the Way of Boolean and Other Lattice-Theoretic Semantices. *Journal of Semantics* 9.2, pp. 95–162. DOI: 10.1093/jos/9.2.95.

Partee, Barbara H. (1989). Possible worlds in model-theoretic semantics: A linguistic perspective. *Possible Worlds in Humanities, Arts and Sciences*. de Gruyter, pp. 93–123. DOI: 10.1515/9783110866858.93.

Partee, Barbara H. and Mats Rooth (1983). Generalized conjunction and type ambiguity. *Meaning, use, and interpretation of language*. Ed. by Rainer Bäuerle, Christoph Schwarze and Arnim von Stechow. Walter de Gruyter, pp. 361–383. DOI: 10.1515/9783110852820.361.

Pinton, Lorenzo (2021). You may like or dislike this thesis, and I do care which: An inquiry into sluicing and free choice. *MSc Thesis, University of Amsterdam*. URL: https://eprints.illc.uva.nl/id/eprint/1801/1/MoL-2021-11.text.pdf.

Portner, Paul (2009). *Modality*. Oxford University Press. DOI: 10.1093/oso/9780199292424.001.0001.

Ronai, Eszter and Ming Xiang (2021). Pragmatic Inferences are QUD-Sensitive: An Experimental Study. *Journal of Linguistics* 57.4, pp. 841–870. DOI: 10.1017/S0022226720000389.

van Rooij, Robert (2000). Permission to Change. *Journal of Semantics* 17.2, pp. 119–143. DOI: 10.1093/jos/17.2.119.

van Rooij, Robert and Katrin Schulz (2007). Only: Meaning and Implicatures. *Questions in dynamic semantics*. Brill, pp. 193–223. DOI: 10.1163/9780080470993_010.

Rooth, Mats and Barbara H. Partee (1982). Conjunction, type ambiguity and wide scope or. *Proceedings of the First West Coast Conference on Formal Linguistics*. Ed. by Daniel P. Flickinger, Marlys Macken and Nancy Wiegand. Vol. 1. Stanford: Stanford Linguistics Association.

Rosella, Giuliano and Jan Sprenger (2024). Causal Modeling Semantics for Counterfactuals with Disjunctive Antecedents. *Annals of Pure and Applied Logic* 175.9, 103336. arXiv: 2304.14817.

Rosen, Gideon (2010). Metaphysical dependence: grounding and reduction. *Modality: metaphysics, logic, and epistemology*. Ed. by Bob Hale and Aviv Hoffmann. Oxford: Oxford University Press, pp. 109–136. DOI: 10.1093/acprof:oso/9780199565818.003.0007.

Ross, Alf (1941). Imperatives and Logic. *Theoria* 7.1, pp. 53–71. DOI: 10.1111/j.1755-2567.1941.tb00034.x.

Rothschild, Daniel and Stephen Yablo (2023). Permissive updates. *Kit Fine on Truthmakers, Relevance, and Non-classical Logic*. Ed. by Federico L. G. Faroldi and Frederik Van De Putte. Springer, pp. 615–662. DOI: 10.1007/978-3-031-29415-0_27.

Santorio, Paolo (2018). Alternatives and truthmakers in conditional semantics. *The Journal of Philosophy* 115.10, pp. 513–549. DOI: 10.5840/jphil20181151030.

Sauerland, Uli (2004). Scalar Implicatures in Complex Sentences. *Linguistics and Philosophy* 27.3, pp. 367–391. DOI: 10.1023/B:LING.0000023378.71748.db.

Schlenker, Philippe (2006). Scopal Independence: A Note on Branching and Wide Scope Readings of Indefinites and Disjunctions. *Journal of Semantics* 23.3, pp. 281–314. DOI: 10.1093/jos/ffl005.

Schulz, Katrin and Robert van Rooij (2006). Pragmatic Meaning and Non-monotonic Reasoning: The Case of Exhaustive Interpretation. *Linguistics and Philosophy* 29.2, pp. 205–250. DOI: 10.1007/s10988-005-3760-4.

Simons, Mandy (2005). Dividing things up: The semantics of *or* and the modal/*or* interaction. *Natural Language Semantics* 13.3, pp. 271–316. DOI: 10.1007/s11050-004-2900-7.

Spector, Benjamin (2003). Scalar implicatures: Exhaustivity and Gricean reasoning. *Proceedings of the Eighth ESSLLI Student Session*. Ed. by Balder ten Cate, pp. 277–288.

— (2007). Scalar implicatures: Exhaustivity and Gricean reasoning. *Questions in dynamic semantics*. Brill, pp. 225–249. DOI: 10.1163/9780080470993_011.

Stalnaker, Robert (1968). A theory of conditionals. *Studies in Logical Theory*. Ed. by Nicholas Rescher. American Philosophical Quarterly Monograph Series 2. Oxford: Blackwell, pp. 98–112. Reprinted in *Ifs: Conditionals, Belief, Decision, Chance and Time*, ed. by William L. Harper, Robert Stalnaker, and Glenn Pearce. Dordrecht: Reidel, 1981, pp. 41–55. DOI: 10.1007/978-94-009-9117-0_2.

Starr, Willow (2016). Expressing permission. *Semantics and Linguistic Theory*. Vol. 26, pp. 325–349. DOI: 10.3765/salt.v26i0.3832.

Strawson, Peter (1950). On referring. *Mind* 59.235, pp. 320–344. DOI: 10.1093/mind/LIX.235.320.

Tieu, Lyn et al. (2016). Children's knowledge of free choice inferences and scalar implicatures. *Journal of Semantics* 33.2, pp. 269–298. DOI: 10.1093/jos/ffv001.

Welker, Katherine A. (1994). Plans in the Common Ground: Toward a Generative Account of Conversational Implicature. PhD thesis. Ohio State University.

Willer, Malte (2018). Simplifying with Free Choice. *Topoi* 37.3, pp. 379–392. DOI: 10.1007/s11245-016-9437-5.

Wilson, Jessica (2012). Fundamental determinables. *Philosophers' Imprint* 12.4, pp. 1–17. URL: http://hdl.handle.net/2027/spo.3521354.0012.004.

Wilson, Jessica (2023). Determinables and Determinates. *The Stanford Encyclopedia of Philosophy*. Ed. by Edward N. Zalta and Uri Nodelman. Spring 2023. Metaphysics Research Lab, Stanford University. URL: https://plato.stanford.edu/archives/spr2023/entries/determinate-determinables/.

Winter, Yoad (2001). *Flexibility Principles in Boolean Semantics. The Interpretation of Coordination, Plurality, and Scope in Natural Language*. Current Studies in Linguistics 37. Cambridge, MA: MIT Press. DOI: 10.7551/mitpress/3034.001.0001.

von Wright, Georg Henrik (1963). *Norm and Action: a logical enquiry*. Routledge & Kegan Paul.

— (1968). Deontic logic and the theory of conditions. *Crítica* 2.6, pp. 3–31. DOI: 10.22201/iifs.18704905e.1968.50.

Yablo, Stephen (2011). A Problem about Permission and Possibility. *Epistemic Modality*. Ed. by Andy Egan and Brian Weatherson. Oxford University Press, pp. 270–294. DOI: 10.1093/acprof:oso/9780199591596.003.0010.

Zondervan, Arjen (2009). Experiments on QUD and Focus as a Contextual Constraint on Scalar Implicature Calculation. *Semantics and Pragmatics: From Experiment to Theory*. Ed. by Uli Sauerland and Kazuko Yatsushiro. Basingstoke: Palgrave Macmillan, pp. 94–112.